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Test I
CS 602/CO434 Image Processing
September, 2024
Marks: 25 Time: 45 minutes

1. a) What is a digital image? Define mathematically. Give a sample example of a digital image according to the definition provided by you. 2
- b) How is a digital image captured? 2
- c) What is meant by image processing? Why are images processed? 2
- d) What are the two main methods for image processing. 2
2. a) Define the power-law transformation process mathematically. Represent it graphically. 2
- b) How is power-law transformation used for gamma correction of computer monitors? 2
3. State whether the following statements are true or false. 1x5=5
 - a) Red light has the shortest wave length in the visible light band. ✓
 - b) Log transformation can be applied to brighten a darker image. -
 - c) image processing operation creates a new image as output. ✓
 - d) Two images of similar size can be added to get a single output image. ✓
 - e) No information loss occurs due to digitation of an image. -
4. a) Name any four of the point processing techniques. 1
- b) What are the left and top neighbours of pixel $f(x, y)$? 1
- c) Provide graphical representation of the histograms of a dark image and a bright image. 1
5. Consider the following image for answering the questions given below.

4	25	18	14
12	15	22	11
7	4	16	28
4	12	14	8

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- a) What is the lowest possible dynamic range? Find the memory size (in bytes) required to store the image with this dynamic range. 1+1=2
- b) Find the negative of the image. $s = L-1-r$ 1
- c) Find the output image if the processing represented by the following graph is applied on the image. 2

